GRIDZILLA

The DSO's monster. Every house keeps it on a leash.

A signed, bounded grid signal for home batteries, cars and heat pumps, with DeepSeek running the grid and a real plant in Blatnica following it live.

Mathew Stafurik · Peter Mikeška · Matthew · Grid-Serving Storage track · github.com/mathewstars/Gridzilla · gridzilla.victronlab.sk

Europe's largest battery fleet is in people's basements. The grid cannot use one of them

0

DSO-level signals to home storage in Slovakia. The only flexibility pilot served the TSO, not the street transformer

4.2 kW

The best case, Germany's §14a: a dimmer for new devices, charging only, used in under 1 % of hours

3 GWh

Home batteries installed in Ukraine in 2025 alone. People charge by hand before the published outage queue

What breaks

- Evening peaks: heat pumps and EV chargers push street transformers past rating; the DSO's only tools are curtailment or copper
- PV noon: everyone exports at once, the feeder voltage rises, new connections get refused or capped for the whole year
- Crisis: when a transformer is hit or the country is short, everyone is cut equally, including the houses that could have ridden through

One signed, bounded signal from the DSO to every house, on whatever path the vendor likes



- The signal names a limit at the grid connection point: import ceiling, export ceiling, export floor, target SoC. It never names a device or a register
- The owner's limits always win: minimum SoC, max current, opt-out. The house may answer partial or refused
- Four-hour maximum, enforced on the platform and again on the device; revert on window end or 120 s without heartbeat
- Pay on the meter: limits are a yes or no per 15-minute interval on the DSO's own smart meter, so there is nothing to game

Six signals, one grammar: limits for a window

PREPARE

Fill the battery, the car and the tank to a target by a deadline, in the quiet hours

ABSORB

PV surplus: cap export, charge, soak it into the car and the hot water, curtail last

SUPPORT

Peak or deficit: "take 3 kW only", and if the street needs more, "give at least 2 kW"

STRETCH · vision

Street is dark: grid-forming inverters island the street, batteries are rationed until restoration, export locked to zero. Today's home inverters cannot form a grid; the signal is ready for the ones that will

RECOVER

After restoration: staggered recharge so the street does not refill at once

RELEASE

Event over. Owner mode, immediately

Envelopes: a SUPPORT or ABSORB carries one ceiling per 15-minute slot, the meter's settlement interval. Cohorts let one signal address half a street, so the other half exports while the first charges.

The household is the unit: everything behind one meter, one agent, eight functions

Battery

the lever of last resort: discharge to cover the house or feed the street, within the owner floor

EV charger

the first lever: throttled 16 to 6 A under a cap, soaks surplus, paused when the street is dark

Heat pump

second lever: tank to 55 °C under surplus, floor at 40 °C under a cap, 3 kW shed per house

Boiler

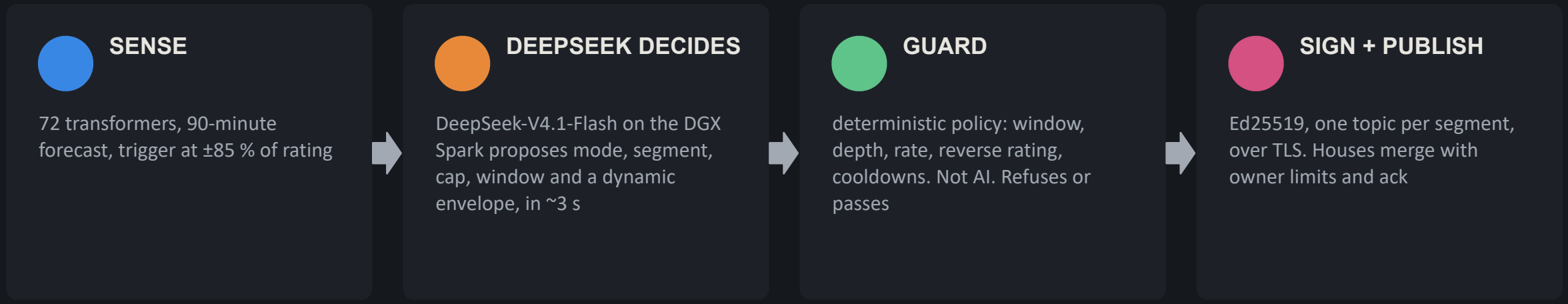
same rules at 65 and 45 °C; an electric boiler is a kilowatt-hour sponge

The agent, any language, forty lines

1. Pin the platform key at registration
2. Subscribe to the segment topics, outbound only
3. Verify signature, sequence, freshness, scope
4. Merge with owner limits: floor, caps, opt-out
5. Apply: the only vendor-specific step
6. Ack, signed with the device key
7. Watch the heartbeat and the window
8. Revert to owner mode

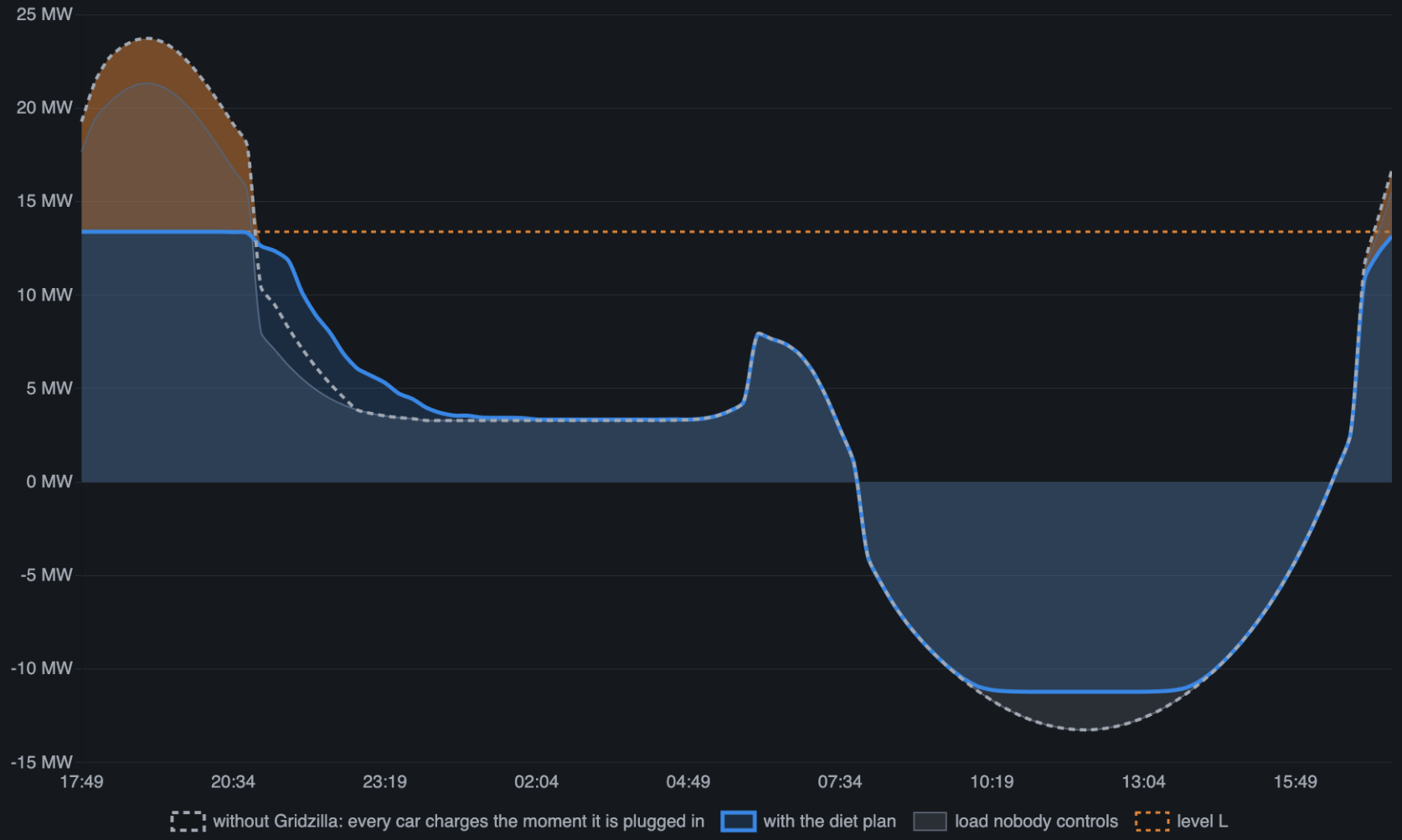
Blatnica, the real one: Victron Cerbo GX with MultiPlus-II and a 15 kWh battery, an 11 kW Victron EV station, and a Viessmann Vitocal heat pump via ViCare. One house on the board.

DeepSeek runs the grid every 15 minutes. The guard is the brake



- Three modes: autonomous, operator approves, off. In autonomous mode nothing waits for a human; the guard and the device-side bounds are the safety
- Scope is physics: a street transformer is relieved only by its own houses; a feeder or substation by every house under it; a village can feed its neighbours through the 22 kV line
- Every decision is logged with its reasoning, latency, verdict and the acks it earned. That log is the audit trail

The monster is on a diet: water-filling flattens every street



79 → 45 %

the planner's forecast for tonight, of transformer rating: 23.7 MW becomes 13.4 MW. Measured result on the next slide

10.5 MWh

of car charging moved from dinner into the night valley; every car full by 07:00

L

the lowest level where the cars fit below it and the peak above it fits in the batteries. Two ceilings per 15 min, per house, signed, renewed every 2 h

What is real, what is modelled, and what the numbers mean

REAL TODAY

- The control path: signed signal in, a Victron plant and a Viessmann heat pump move, verified acks come back
- The flattening algorithm and the guard, running on the fleet every two grid hours
- Measured on streets with cars: 18-31 % off the evening peak, inside what published smart-charging and home-battery pilots report (3-6 kW per EV house, 1-2 kW per battery at the peak)

MODELLED

- The fleet: 271 simulated households, each standing for 8 real ones: 2,184 program houses with 10-25 kWh batteries, 1,190 with an 11 kW charger needing 6-26 kWh a night
- The "without" curve is the same houses with no signal: a car charges on 55 % of nights, arrivals spread over three hours, full power until full. No inflated coincidence
- Program houses carry ~30 % of a street's evening load here: a 2030 picture, not a street today

−176 kW

one street, Speising-22: 574 → 398 kW at the evening peak, 31 %, measured tonight

−5.2 MW

Vienna with 2,184 enrolled houses: 26.6 → 21.7 MW at the evening peak, 19 %, measured tonight

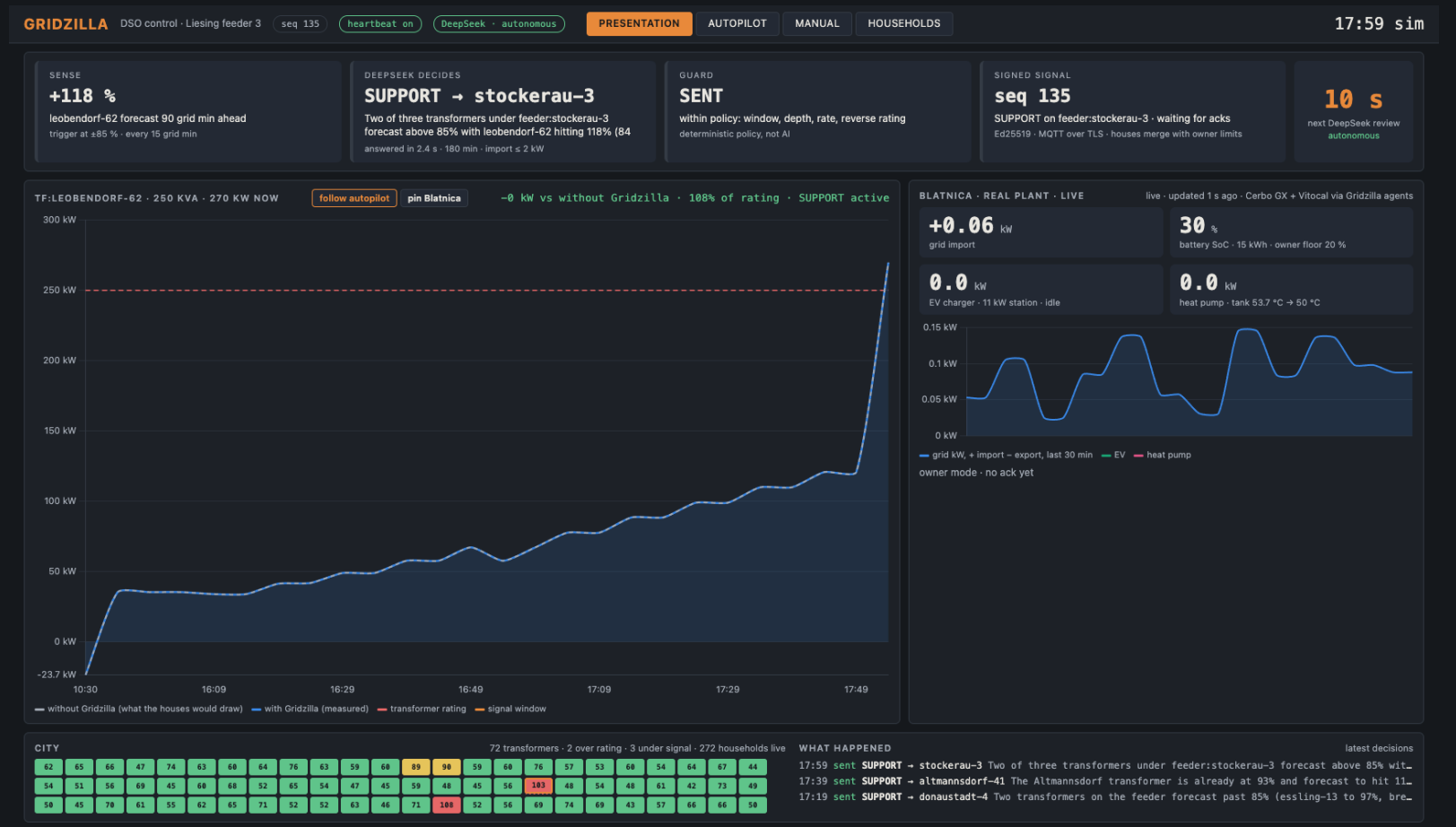
0 kW

on a street where no car charged tonight: the gap is the cars and the batteries, nothing else

1

real house in the fleet tonight. The fleet is the part we are asking you to build

Live, three windows: this deck, the Gridzilla board, VRM



Four minutes, alt-tab

- Board: city view, 72 transformers, 272 households live, one of them real
- Board: one DeepSeek review, sense, decide, guard, signed signal, countdown
- Board: the curve, with vs without, on the transformer it chose
- Board, Flattening tab: the 24 h diet plan for Vienna, then one street; cars moved into the valley, batteries eat the peak
- Board, pin Blatnica: send SUPPORT from the Manual tab; tiles move within seconds
- VRM: the same plant from Victron's side, grid setpoint and battery power reacting
- Board: pause the heartbeat, every house reverts to owner mode

Everything on that screen runs on real infrastructure today

272

households on the board, 8,070 houses represented

72

street transformers, 24 feeders, 8 substations with Wiener Netze names

2

real devices: a Victron plant and a Viessmann heat pump, 60 km from here

~3 s

DeepSeek decision latency on the DGX Spark

Real plant

Cerbo GX on Venus OS Large, a 53-node Node-RED agent we deploy over the admin API; EV Charging Station throttled first; ESS limits written and restored

Heat pump

Vitocal CU401B through the ViCare REST API with a refresh token; hot-water setpoint and one-time charge, verified on the unit

Transport

Mosquitto on our own server, Let's Encrypt, TLS only, credentials and topic access lists; devices open outbound connections only

Platform

FastAPI on a Ubuntu box behind Apache and Let's Encrypt; registry tree, guard, signing, history, autopilot; public board at gridzilla.victronlab.sk

Simulation

262 households with battery, PV, EV commutes, heat pumps and boilers with real tank physics; each agent speaks the same protocol as the Cerbo

Protocol

Frozen as PROTOCOL.md with JSON Schemas for envelope, signal and ack; any vendor can implement the eight functions from it

Any vendor, four lines each. Any household, paid on the meter

Three paths to the same agent

- Vendor cloud: GoodWe SEMS, Huawei FusionSolar, SolarEdge, Tesla Fleet, SMA, Fronius. Export limit, discharge to cover load, forced discharge schedule, grid charge schedule
- Local controller: Home Assistant, EVCC, OpenEMS, SunSpec Modbus. What the demo runs on Node-RED today
- Standards gateway: EEBus LPC and LPP are literally our import and export caps; OpenADR 3, IEC 61850, OCPP 2.0.1 for chargers
- GoodWe is one apply function away: the registers and the hardware are on the team

Why a DSO pays and a household joins

- Limits are verifiable, reductions are not: the DSO's own smart meter answers yes or no per 15-minute interval. No baseline to inflate
- No ack, no pay; compliance gates the perks; the DSO's existing disconnect is the backstop in a crisis
- The hook: a dynamic export limit beats today's static cap on all but a few afternoons, and gets a refused connection approved
- Plus an availability fee, per-kWh payment for forced feed-in, a full battery before an outage, and a shorter shedding queue
- Noon export is worthless when everyone exports; the same kilowatt-hours leave at six, paid

Production at a DSO: the effort, the cost, the saving

EFFORT

- Protocol and agents are the small part; they exist
- The DSO must add: enrolment bound to the meter point and operator login, a PKI with key rotation, settlement on meter data, the real network model feeding the tree and ratings, 24/7 operations, NIS2 and energy-law compliance
- 10-house pilot on one transformer: 3 months
- A district of a few thousand homes: 12-18 months, 6-8 people, roughly 1.5-3 M€ including the security audit

COST TO RUN

- The platform is cheap: two servers and a broker
- The real cost is the incentive: a flat 100-200 € per enrolled house per year, plus energy at crisis prices during STRETCH and SUPPORT
- Paid only to houses whose meter proves they followed the signal. No enforcement, no payment, no saving

SAVING

- Street transformer upgrade 50-150 k€, feeder reinforcement hundreds of k€, a substation tens of M€
- Every street held at ~70 % at the evening peak defers the reinforcement the EV and heat-pump wave otherwise forces this decade
- Vienna scale: hundreds of M€ of capex deferred against a few M€ a year of incentives
- Plus a tool for the outage: PREPARE before it, RECOVER after it, and with grid-forming inverters STRETCH keeps a dark street alive from its own batteries

NEXT

A pilot street, a second vendor, a DSO that wants the signal

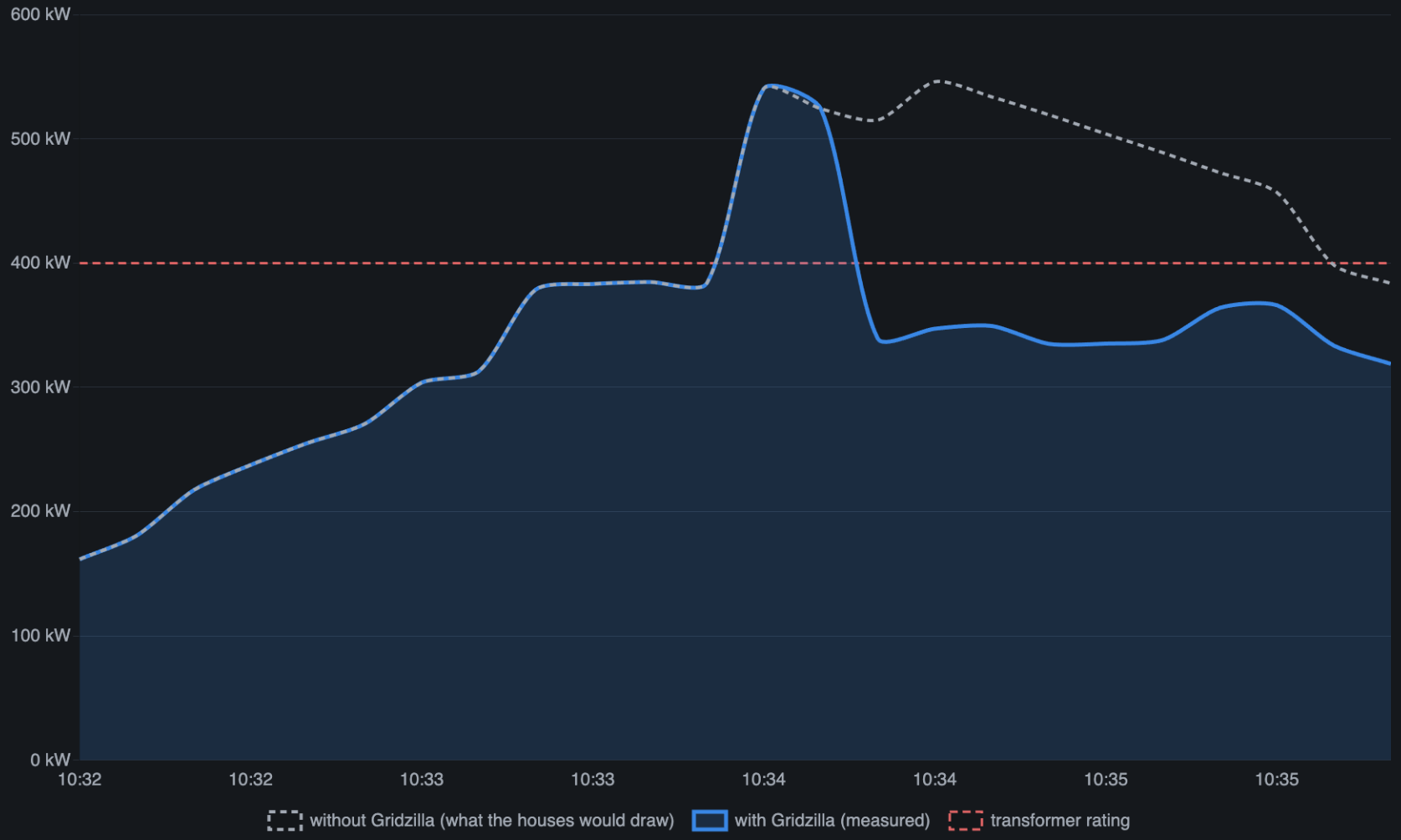
Enrolment bound to the meter point and operator login, two weeks. GoodWe agent on our own plant, one week.

A ten-house pilot with a Slovak or Austrian DSO on one real transformer, this winter.

The vision: streets that island on grid-forming inverters when the 22 kV line is gone, rationed by the same signal.

github.com/mathewstars/Gridzilla · gridzilla.victronlab.sk · protocol, platform, agents and this deck are in the repo

What the flattening looks like: speising-22, 400 kVA, from an earlier run



−199 kW

largest cut below the counterfactual on this transformer

2 kW

import cap per house DeepSeek chose; EVs throttled first, batteries second

0

owner floors crossed; partial acks where a battery hit its reserve

Security of the control plane: the DSO can ask, it can never command

Authenticity

every signal, heartbeat and ack carries an Ed25519 signature; the device pins the platform key and signs with its own

Replay

monotonic sequence numbers persisted on the device, 120 s freshness, heartbeats with sequence and timestamp

Bounded

limits, never setpoints; four-hour window enforced on the device itself; depth and rate capped by the guard

Owner veto

minimum SoC, max current, opt-out merged on the device; a compromised DSO key moves nothing past them

Fail-safe

window end, heartbeat loss or RELEASE restore the owner's settings; a dead platform leaves every house in owner mode

Transport

TLS with a real certificate, per-role credentials, topic ACLs; a device cannot publish a signal even if it tried

Named before you ask

Enrolment is open, any device id can register a key; production binds it to the meter point with a one-time code. The operator board has no login for the demo; production is SSO plus a two-person rule for feeder-level signals. Neither changes the protocol.